Aaron Leevord Fernandes

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EDUCATION

Indiana University Bloomington Aug 2023 - May 2025
Master of Science in Intelligent Systems Engineering - Computer Engineering ***GPA: 3.9/4.0***

Goa University Aug 2017 - Jul 2021
Bachelor of Engineering - Electronics and Telecommunication ***GPA: 3.7/4.0***

PROJECTS

Visual Language Model (VLM) from Scratch | PyTorch, Vision Transformer, CUDA July 2025 – Oct 2025

- Built **PolyGama Visual Language Model** from scratch, combining a **ViT encoder** with a Transformer language model.
- Implemented **contrastive** image–text embeddings, rotary positional encodings **(RoPE)**, and custom attention layers, enabling conditional text generation from images.
- Optimized for **GPU acceleration** using **CUDA** mixed-precision training, asynchronous data transfer, and **Flash attention kernels**, achieving 3× faster inference and 40% lower memory usage.

Vision Transformer-Based Unsupervised Anomaly Detection on Industrial Dataset Dec 2024 - Jan 2025

- Designed a **ViT-based autoencoder** with **capsule networks** and **Gaussian Mixture density modeling** for **industrial anomaly detection**, achieving **20% AUROC** improvement over CNN baselines.
- Optimized inference pipeline in PyTorch for large-scale datasets, demonstrating scalable unsupervised modeling strategies relevant to personalization and **anomaly detection** at enterprise scale.

Booking Management System | Next.js, Python, FastAPI, MySQL June 24 - Aug 2024

- Built **10+ key features** (secure login, email, chat, Stripe) with unit tests, reducing production bugs by 40%.
- Implemented **RESTful APIs** for bookings and payments, integrating Stripe and optimizing frontend–backend communication, improving UX and transaction success rates.

Efficient Model-Based Deep Reinforcement Learning With Predictive Control Aug 2024 - Dec 2024

- Developed an **MPC-driven RL framework** with world models and trajectory optimization, converging 5× faster than model-free baselines.
- Improved sample efficiency and inference performance in PyTorch, focusing on efficient **GPU-accelerated** training and inference pipelines.

Predictive Modeling for NCAA Women's Basketball (Random Forests) Sep 2024 - Oct 2024

- Built **predictive Random Forest models** on large-scale customer behavior data, improving accuracy by **19%** and ranking **top 5/67** teams in national competition. And delivered business insights that projected up to **15% revenue growth** and **10% fan engagement gains**.

EXPERIENCE

Machine Learning Engineer, Image Processing, Computer Vision and Optics (Leung Research Group) Apr 2024 – Present

Biomedical Video Segmentation Project

- Designed deep learning **CNN-GRU and Transformers** based architectures in Pytorch for Optical Coherence Tomography (OCT) **unsupervised video segmentation** and **spatio-temporal analysis** to enable virtual histology (tissue analysis)
- Combined **Fourier analysis**, and deep learning, providing 50% better details than previous works.
- Solved tissue drift and warping during imaging by classical **affine registration** methods (Dipy) and voxel morph inspired U-Net based architectures for regularized **diffeomorphic registration** decreasing MSE by 10%.

Metasurfaces: High-Q Biomedical sensing Project

- Built **High Performance Computing** (Slurm, MPI) accelerated Lumerical FDTD simualtion pipelines for Meta-Surface design and **synthetic data generation** improving generation speeds by 80%.
- Engineered **U-Net** based FDTD solver **surrogate models** for forward and inverse metasurface design in **tensorflow**, reducing simulation runtime by 70% while maintaining 95% accuracy, achieving up to 15% gains over prior benchmarks.

Software Engineer, Machine Learning (Indiana University) Jan 2025 - May 2025

Multi Target Path Planning and Routing for Robotic Environments Project

- Identified limitations of value-based methods (e.g., DQN: 20% success) in stochastic, multi-target routing environments.
- Implemented and benchmarked **policy-gradient** algorithms **(REINFORCE, PPO, TRPO, SAC, A2C)** in PyTorch across custom tabular path-planning tasks with moving targets and obstacles; **PPO** achieved up to 90% success.
- Addressed **sparse reward problem** by integrating **Goal-Conditioned** Actor-Critic with **Stein Variational Goal Generation** and **Hindsight Experience Replay (HER)**, enabling successful completion of complex multi-goal tasks **previously unsolvable** under standard RL setups.

Software Engineer (Tata Consultancy Services - HDFC Bank) Jan 2022 - Jan 2023

- Developed robust financial transaction systems using **Java, Spring and Hibernate**, improving processing efficiency
- Engineered SOAP/HTTP APIs and **Microservices**, enhancing system interoperability and reducing latency by 10-15%
- Streamlined transactional workflows, ensuring data consistency and seamless integration, cutting reconciliation errors 7–15%

TECHNICAL SKILLS

Machine Learning / AI: CUDA, Triton, PyTorch, TensorFlow, Keras, JAX, LangChain, LangGraph, spaCy, NLTK, RAG, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, FAISS,
Systems / Cloud / DevOps: AWS (SageMaker, EC2, Lambda, S3, RDS, CloudFormation), GCP, Docker, Kubernetes, Spark, Hadoop, Microservices
Programming Languages: Python, Java, C, C++, JavaScript (Node.js/TypeScript), SQL
Full-Stack Development: React.js, Next.js, Node.js, Spring Boot, Hibernate, PostgreSQL, MongoDB, Redis, REST, GraphQL, Neo4j
Certifications: AWS Developer Associate, AWS Cloud Practitioner